DATA ARCHITECTURE DOCUMENT

Entity-Relationship Diagrams

Complete Database Schema Reference — All 6 Microservice Databases

Version: 2.0 | **Date:** April 2026

Databases: 6 SQL Server Instances (Database-per-Service)

ORM: Entity Framework Core — Code-First Migrations

Total Entities: 28 tables across all bounded contexts

Classification: Confidential — Internal Use Only

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1. Data Architecture Overview

> **Design Principle:** Each microservice owns a dedicated SQL Server database. No direct cross-database joins or shared schemas. Inter-service data access is achieved exclusively through published REST APIs or asynchronous domain events via RabbitMQ.

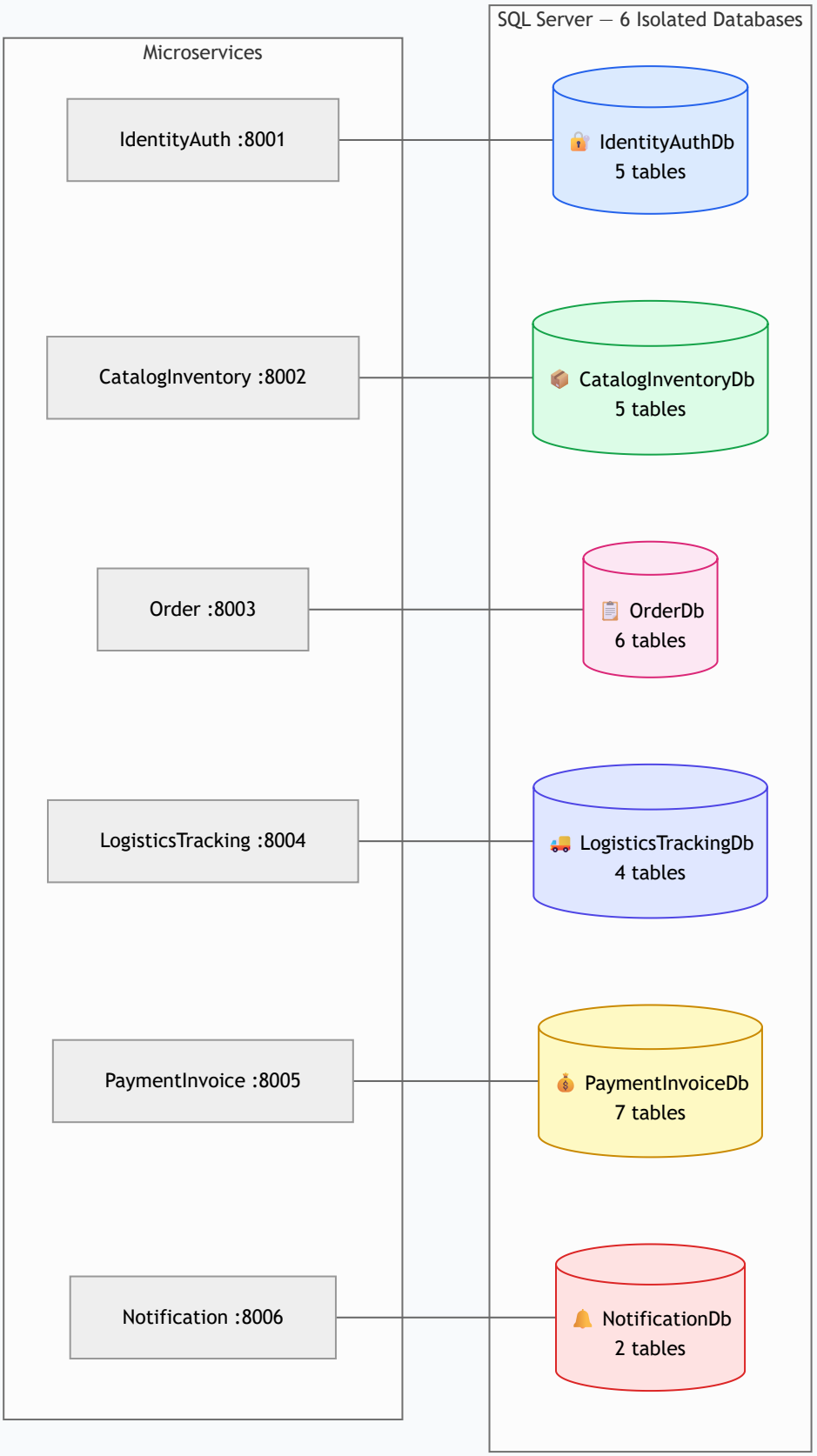

FIGURE 1 — DATABASE-PER-SERVICE OWNERSHIP MODEL (28 TOTAL TABLES)

2. Cross-Service Data Flow Map

While databases are isolated, services reference each other's entities by ID. This diagram shows how primary keys are referenced across boundaries:

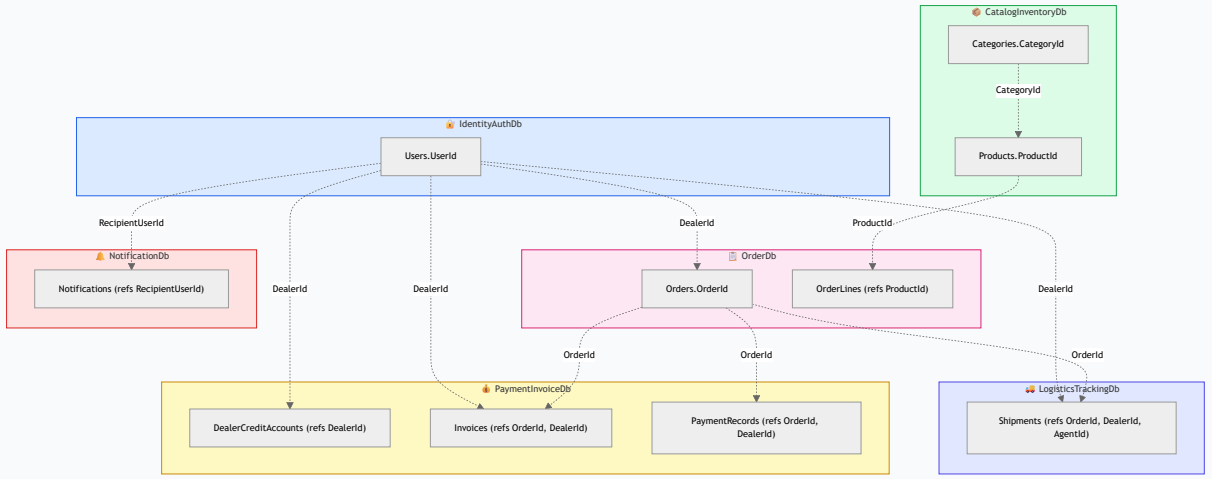

FIGURE 2 — CROSS-SERVICE ID REFERENCE MAP (LOGICAL, NOT PHYSICAL FK)

> **Important:** Dashed arrows represent *logical references* by ID value — NOT physical foreign keys. Each service stores only the GUID/ID of entities from other bounded contexts. Data consistency is maintained through the Outbox + RabbitMQ event pattern.

3. IdentityAuth Database

IdentityAuthDb

5 Tables · Port 8001 · User Lifecycle & Authentication

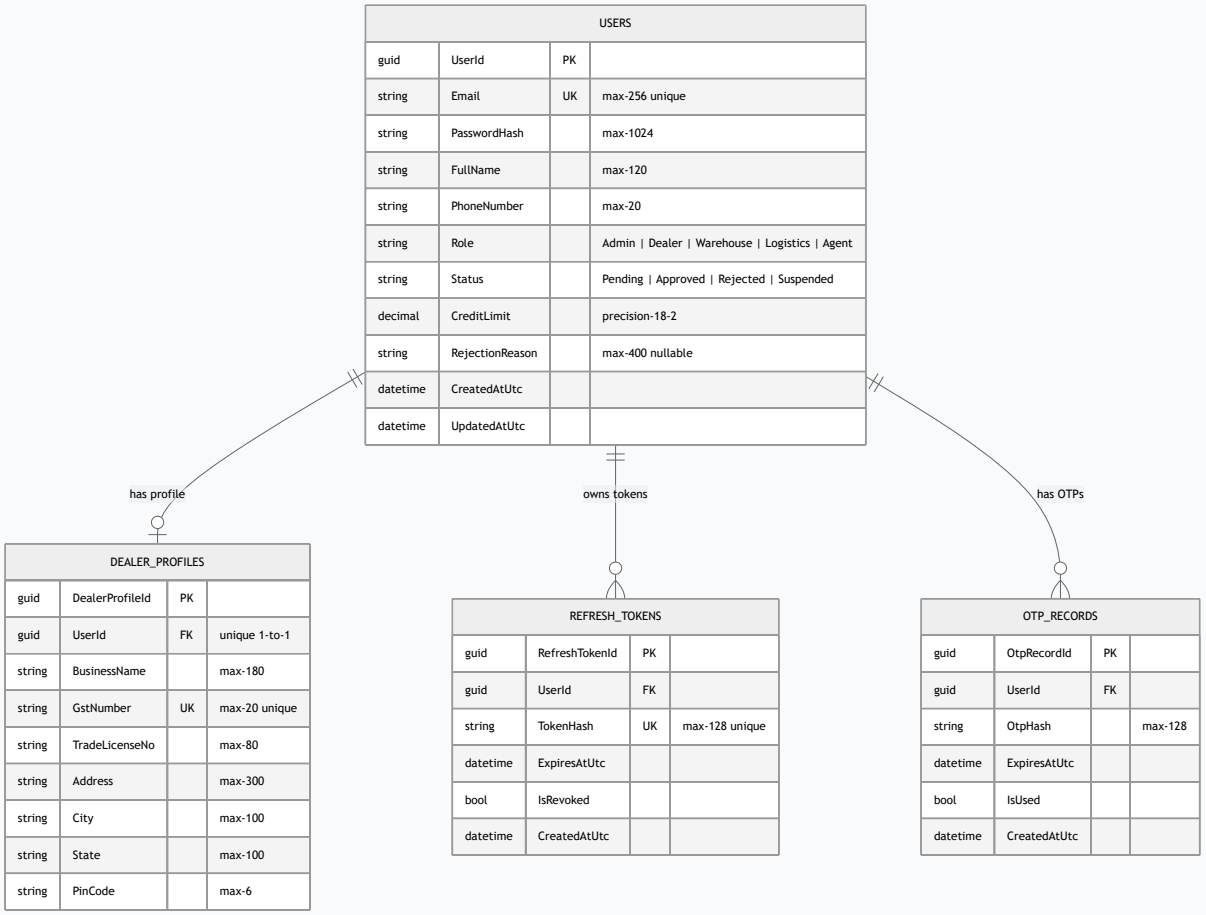

FIGURE 3 — IDENTITYAUTH ER DIAGRAM (4 DOMAIN TABLES + OUTBOX)

3.1 Column Details & Constraints

TABLE	CONSTRAINT	TYPE	DETAILS
Users	Email	Unique Index	Ensures no duplicate registrations
Users	(Role, Status, CreatedAtUtc)	Composite Index	Optimizes admin dealer list queries with filtering
DealerProfiles	GstNumber	Unique Index	GST compliance — one profile per GST number
DealerProfiles	UserId	FK + Unique	1:1 relationship with Users, CASCADE delete
RefreshTokens	TokenHash	Unique Index	Fast lookup for token refresh/revocation
RefreshTokens	UserId	FK	N:1 to Users, CASCADE delete
OtpRecords	UserId	FK	N:1 to Users, CASCADE delete

4. CatalogInventory Database

CatalogInventoryDb

5 Tables · Port 8002 · Products, Categories, Stock & Reviews

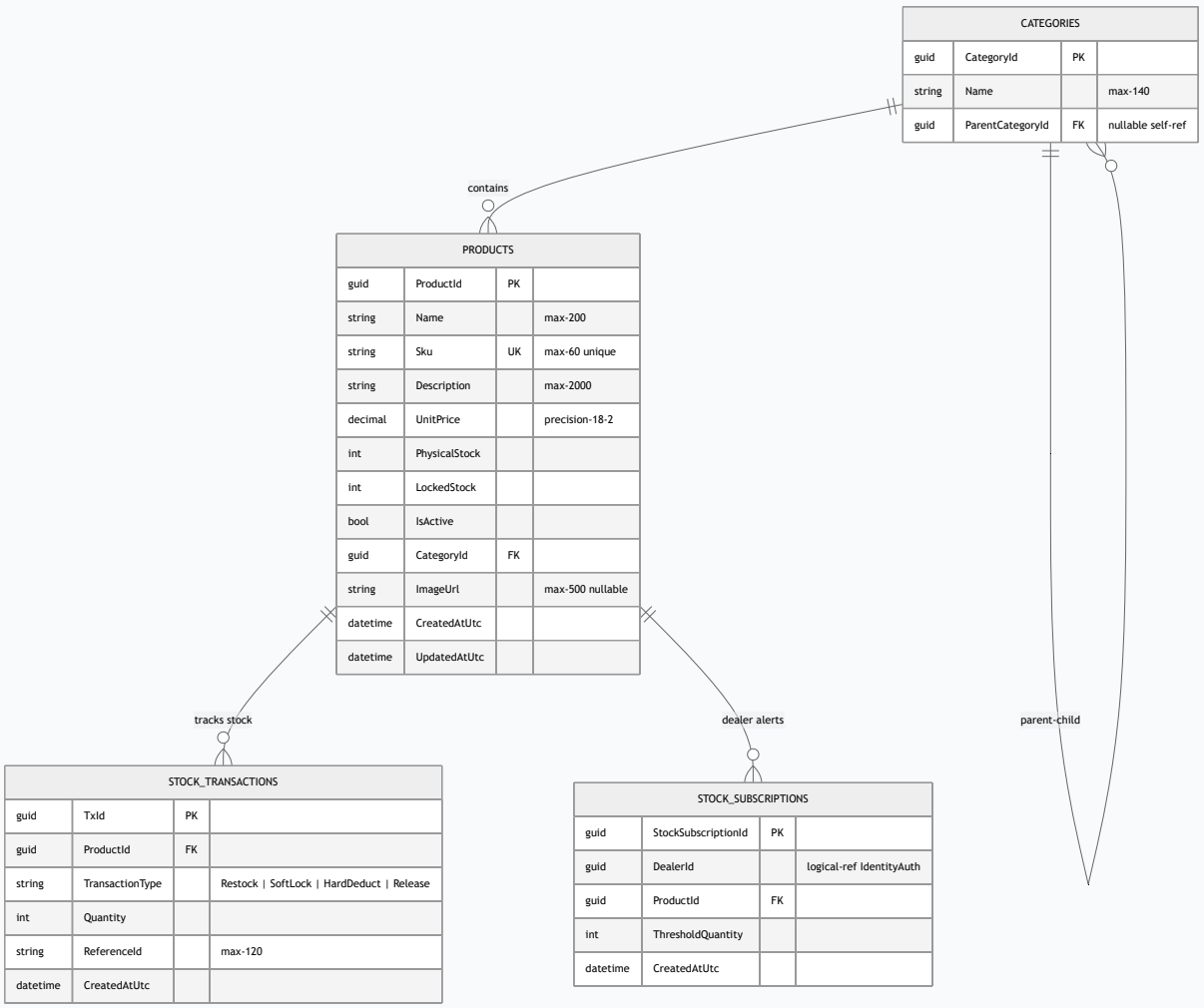

FIGURE 4 — CATALOGINVENTORY ER DIAGRAM (4 DOMAIN TABLES + OUTBOX)

4.1 Column Details & Constraints

TABLE	CONSTRAINT	TYPE	DETAILS
Products	Sku	Unique Index	Business key — unique product identifier
Products	CategoryId	FK	N:1 to Categories, RESTRICT delete (preserve products)
Products	AvailableStock	Computed	PhysicalStock - LockedStock (ignored by EF, computed in domain)
Categories	ParentCategoryId	Self-referencing FK	Nullable — root categories have NULL parent, RESTRICT delete
StockTransactions	ProductId	FK	N:1 to Products, CASCADE delete
StockSubscriptions	(DealerId, ProductId)	Composite Unique	One subscription per dealer per product

Stock Calculation: `AvailableStock = PhysicalStock - LockedStock`. This is a computed property in the domain model (EF `Ignore`), not a persisted column. Stock changes flow through StockTransactions for full audit trail.

5. Order Database

OrderDb

6 Tables · Port 8003 · Order Aggregate, Saga State & Returns

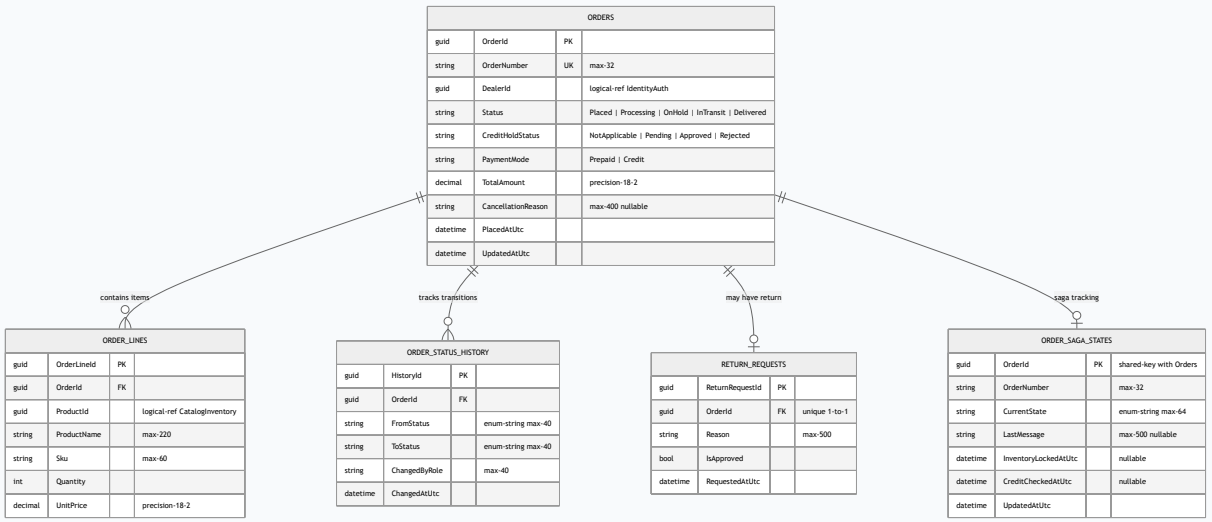

FIGURE 5 — ORDER ER DIAGRAM (5 DOMAIN TABLES + OUTBOX)

5.1 Column Details & Constraints

TABLE	CONSTRAINT	TYPE	DETAILS
Orders	OrderNumber	Unique Index	Human-readable order reference (ORD-YYYYMMDD-XXXX)
Orders	(DealerId, PlacedAtUtc)	Composite Index	Optimizes "my orders" dealer queries
Orders	(Status, PlacedAtUtc)	Composite Index	Optimizes admin order list with status filtering
OrderLines	OrderId	FK	N:1 to Orders, CASCADE delete
OrderLines	LineTotal	Computed	Quantity × UnitPrice (EF Ignore)
OrderStatusHistory	OrderId	FK	N:1 to Orders, CASCADE delete
ReturnRequests	OrderId	FK + Unique	1:1 with Orders, CASCADE delete
OrderSagaStates	OrderId	PK + FK	1:1 with Orders — uses OrderId as PK

6. LogisticsTracking Database

LogisticsTrackingDb

4 Tables · Port 8004 · Shipments, Events, Ops State & Agent Assignment

SHIPMENTS			
guid	ShipmentId	PK	
string	ShipmentNumber	UK	max-32 unique
guid	OrderId		logical-ref OrderDb
guid	DealerId		logical-ref IdentityAuth
guid	AssignedAgentId		nullable logical-ref
string	Status		Created | Assigned | PickedUp | InTransit | Delivered
string	DeliveryAddress		max-500
string	City		max-100
string	State		max-100
string	PostalCode		max-12
string	VehicleNumber		max-32 nullable
string	AssignmentDecisionStatus		Pending | Accepted | Rejected
string	AssignmentDecisionReason		max-500 nullable
datetime	AssignmentDecisionAtUtc		nullable
int	DeliveryAgentRating		nullable 1-5 scale
string	DeliveryAgentRatingComment		max-500 nullable
datetime	DeliveryAgentRatedAtUtc		nullable
guid	DeliveryAgentRatedByUserId		nullable
datetime	CreatedAtUtc		
datetime	UpdatedAtUtc		

timeline

ops tracking

SHIPMENT_EVENTS			
guid	ShipmentEventId	PK	
guid	ShipmentId	FK	
string	Status		enum-string max-32
string	Note		max-500
string	UpdatedByRole		max-40
datetime	OccurredAtUtc		

SHIPMENT_OPS_STATES			
guid	ShipmentId	PK	shared-key with Shipments
string	HandoverState		Ready | Pending | Exception
string	HandoverExceptionReason		max-300 nullable
bool	RetryRequired		
int	RetryCount		
string	RetryReason		max-300 nullable
datetime	UpdatedAtUtc		

FIGURE 6 — LOGISTICSTRACKING ER DIAGRAM (3 DOMAIN TABLES + OUTBOX)

6.1 Column Details & Constraints

TABLE	CONSTRAINT	TYPE	DETAILS
Shipments	ShipmentNumber	Unique Index	Human-readable shipment reference
Shipments	(DealerId, CreatedAtUtc)	Composite Index	Dealer's shipment history queries
Shipments	(AssignedAgentId, CreatedAtUtc)	Composite Index	Agent's assigned shipments view
Shipments	CreatedAtUtc	Index	Time-based shipment listing (admin)
ShipmentEvents	ShipmentId	FK	N:1 to Shipments, CASCADE delete
ShipmentOpsStates	ShipmentId	PK + FK	1:1 with Shipments — uses ShipmentId as PK, CASCADE delete

7. PaymentInvoice Database

PaymentInvoiceDb

7 Tables · Port 8005 · Credit, Invoicing, Workflow & Payments

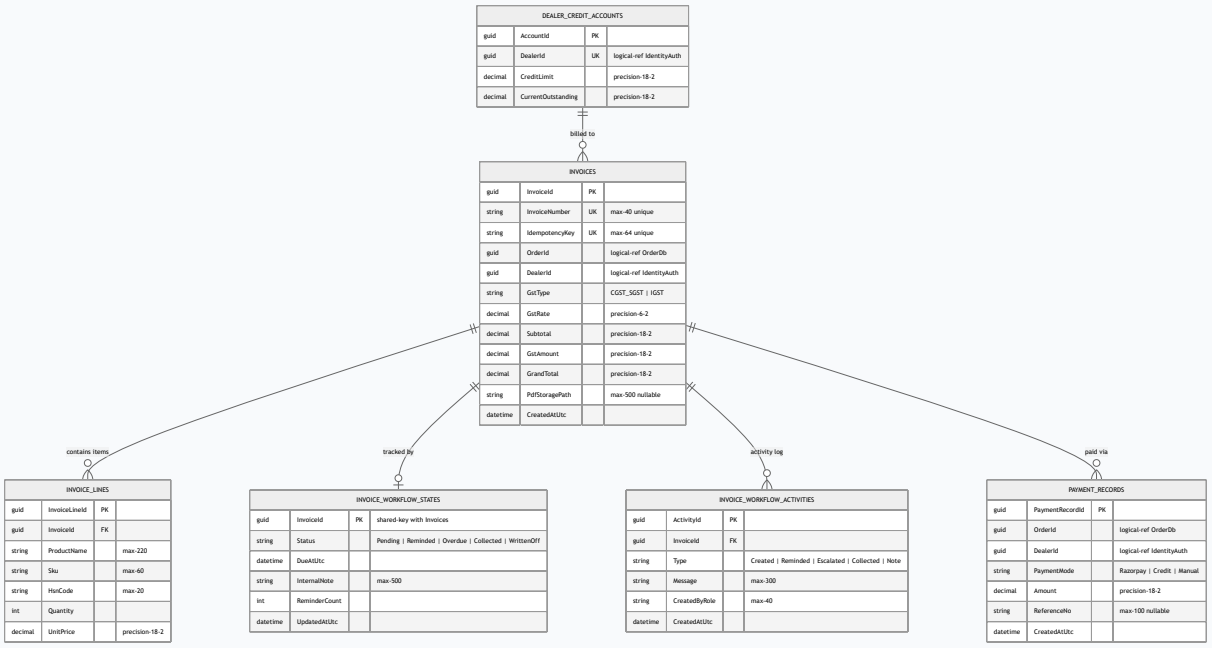

FIGURE 7 — PAYMENTINVOICE ER DIAGRAM (6 DOMAIN TABLES + OUTBOX)

7.1 Column Details & Constraints

TABLE	CONSTRAINT	TYPE	DETAILS
DealerCreditAccounts	DealerId	Unique Index	One credit account per dealer
DealerCreditAccounts	AvailableCredit	Computed	CreditLimit - CurrentOutstanding (EF Ignore)
Invoices	InvoiceNumber	Unique Index	Human-readable invoice ID (INV-YYYYMMDD-XXXX)
Invoices	IdempotencyKey	Unique Index	Prevents duplicate invoice creation
Invoices	(DealerId, CreatedAtUtc)	Composite Index	Dealer's invoice listing
InvoiceLines	InvoiceId	FK	N:1 to Invoices, CASCADE delete
InvoiceLines	LineTotal	Computed	Quantity × UnitPrice (EF Ignore)
InvoiceWorkflowStates	InvoiceId	PK + FK	1:1 with Invoices, CASCADE delete
InvoiceWorkflowActivities	(InvoiceId, CreatedAtUtc)	Composite Index	Chronological activity timeline
InvoiceWorkflowActivities	InvoiceId	FK	N:1 to Invoices, CASCADE delete

8. Notification Database

NotificationDb

2 Tables · Port 8006 · Multi-Channel Notification Feed

NOTIFICATIONS			
guid	NotificationId	PK	
guid	RecipientUserId		logical-ref IdentityAuth
string	Title		max-180
string	Body		max-4000
string	SourceService		max-100
string	EventType		max-100
string	Channel		InApp | Email
string	Status		Pending | Sent | Failed | Read
string	FailureReason		max-1000 nullable
bool	IsRead		
datetime	CreatedAtUtc		
datetime	SentAtUtc		nullable
datetime	ReadAtUtc		nullable

FIGURE 8 — NOTIFICATION ER DIAGRAM (1 DOMAIN TABLE + OUTBOX)

8.1 Column Details & Constraints

TABLE	CONSTRAINT	TYPE	DETAILS
Notifications	RecipientUserId	Index	Fast lookup for user's notification feed
Notifications	CreatedAtUtc	Index	Admin notification listing (chronological)
Notifications	(RecipientUserId, CreatedAtUtc)	Composite Index	Optimized "my notifications" query with time ordering

9. Shared Outbox Pattern Schema

Every microservice includes an identical `OutboxMessages` table for guaranteed event delivery:

OUTBOX_MESSAGES			
guid	MessageId	PK	
string	EventType		max-200
string	Payload		JSON text required
string	Status		Pending | Published | Failed
datetime	CreatedAtUtc		
datetime	PublishedAtUtc		nullable
int	RetryCount		
string	Error		max-2000 nullable

FIGURE 9 — OUTBOX MESSAGE SCHEMA (PRESENT IN ALL 6 DATABASES)

FIELD	TYPE	MAX LENGTH	PURPOSE
MessageId	GUID (PK)	—	Unique identifier for idempotent publish
EventType	string	200	Fully qualified event name (e.g., OrderPlaced)
Payload	nvarchar(max)	Unlimited	JSON-serialized event payload
Status	enum → string	32	Pending → Published by dispatcher, or Failed after max retries
CreatedAtUtc	datetime	—	Written in same transaction as domain data
PublishedAtUtc	datetime?	—	When successfully published to RabbitMQ
RetryCount	int	—	Number of publish attempts by background worker
Error	string?	2000	Last error message for failed publishes

10. Indexing Strategy & Performance

All indexes are defined in EF Core `OnModelCreating` and applied via code-first migrations. Additional operational indexes are managed via the SQL patch at `scripts/migrations/IndexingPatch.sql`.

DATABASE	TABLE	INDEX COLUMNS	TYPE	RATIONALE
`IdentityAuth`	Users	(Email)	Unique	Login lookup, duplicate prevention
`IdentityAuth`	Users	(Role, Status, CreatedAtUtc)	Composite	Admin dealer listing with role/status filter
`IdentityAuth`	DealerProfiles	(GstNumber)	Unique	GST compliance uniqueness
`IdentityAuth`	RefreshTokens	(TokenHash)	Unique	Token refresh/revocation lookup
`CatalogInventory`	Products	(Sku)	Unique	Product search by SKU
`CatalogInventory`	StockSubscriptions	(DealerId, ProductId)	Composite Unique	One sub per dealer per product
`Order`	Orders	(OrderNumber)	Unique	Order lookup by business reference
`Order`	Orders	(DealerId, PlacedAtUtc)	Composite	"My orders" dealer query
`Order`	Orders	(Status, PlacedAtUtc)	Composite	Admin filtered order listing
`Logistics`	Shipments	(ShipmentNumber)	Unique	Shipment lookup
`Logistics`	Shipments	(DealerId, CreatedAtUtc)	Composite	Dealer shipment history
`Logistics`	Shipments	(AssignedAgentId, CreatedAtUtc)	Composite	Agent workload view
`Payment`	DealerCreditAccounts	(DealerId)	Unique	One credit account per dealer
`Payment`	Invoices	(InvoiceNumber)	Unique	Invoice lookup
`Payment`	Invoices	(IdempotencyKey)	Unique	Duplicate invoice prevention
`Payment`	Invoices	(DealerId, CreatedAtUtc)	Composite	Dealer invoice listing
`Notification`	Notifications	(RecipientUserId, CreatedAtUtc)	Composite	User notification feed

11. Data Integrity & Constraints Summary

11.1 Relationship Cardinality Summary

RELATIONSHIP	CARDINALITY	DELETE BEHAVIOR
Users → DealerProfile	1:1	CASCADE
Users → RefreshTokens	1:N	CASCADE
Users → OtpRecords	1:N	CASCADE
Categories → Products	1:N	RESTRICT
Categories → Children	1:N (self-ref)	RESTRICT
Products → StockTransactions	1:N	CASCADE
Products → StockSubscriptions	1:N	CASCADE
Orders → OrderLines	1:N	CASCADE
Orders → StatusHistory	1:N	CASCADE
Orders → ReturnRequest	1:1	CASCADE
Orders → OrderSagaStates	1:1	PK sharing
Shipments → ShipmentEvents	1:N	CASCADE
Shipments → ShipmentOpsStates	1:1	CASCADE
Invoices → InvoiceLines	1:N	CASCADE
Invoices → WorkflowState	1:1	CASCADE
Invoices → WorkflowActivities	1:N	CASCADE

11.2 Computed Properties (EF Ignore)

ENTITY	PROPERTY	FORMULA
Products	AvailableStock	PhysicalStock - LockedStock
OrderLines	LineTotal	Quantity × UnitPrice
InvoiceLines	LineTotal	Quantity × UnitPrice
DealerCreditAccounts	AvailableCredit	CreditLimit - CurrentOutstanding

11.3 Enum Storage Convention

> **Convention:** All enum fields are stored as `string` (not int) using EF Core `HasConversion<string>()`. This improves query readability in SQL Server Management Studio and avoids breaking changes when enum values are reordered.

Supply Chain Management Platform — Entity-Relationship Diagrams v2.0

28 Tables · 6 Databases · Confidential · April 2026